
Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

The normalized difference vegetation index (NDVI) is a simple indicator of green vegetation that can be used to analyse remote sensing measurements. The NDVI value depends on the RED and NIR component of the reflected light. Which of the following formula shows the relationship between NDVI, RED and INR?

- A. $NDVI = (NIR-RED)/(NIR+RED)$
- B. $NDVI = (2 * NIR)/(NIR+RED)$
- C. $NDVI = (2 * RED)/(NIR-RED)$
- D. $NDVI = (NIR+RED)/(NIR-RED)$

Answer: A

Detailed Solution:

$$NDVI = (NIR-RED)/(NIR+RED)$$

QUESTION 2

To predict the PM_{2.5} atmospheric air pollution Muthukumar et. al. have used some advanced deep learning models with remote-sensing satellite imagery of air pollution, ground-based sensors monitoring data of air pollutants and meteorological features. Which of the following models are used in this work?

[Ref: Muthukumar P, Cocom E, Nagrecha K, Comer D, Burga I, Taub J, Calvert CF, Holm J, Pourhomayoun M. Predicting PM_{2.5} atmospheric air pollution using deep learning with meteorological data and ground-based observations and remote-sensing satellite big data. *Air Quality, Atmosphere & Health*. 2021 Nov 23:1-4.]

- A. GCN and LSTM
- B. GCN and ConvLSTM
- C. CNN and ConvLSTM
- D. CNN and LSTM

Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

Answer: B

Detailed Solution:

Graph Convolutional Network (GCN) and Convolutional Long Short-Term Memory are the models used in this work.

QUESTION 3

Which of the following is a key difference between a Gated Recurrent Unit (GRU) and a Long Short-Term Memory (LSTM) network?

- A. GRUs have fewer gates and fewer parameters than LSTMs.
- B. LSTMs do not use gating mechanisms, whereas GRUs use update and reset gates.
- C. GRUs are designed exclusively for image processing, whereas LSTMs are designed exclusively for natural language processing.
- D. LSTMs cannot capture long-term temporal dependencies, whereas GRUs can capture arbitrarily long dependencies.

Answer: A

Detailed Solution:

A simple GRU contains two gates (reset gate and update gate). An LSTM relies on three distinct gates to control its internal state: the input gate, forget gate, and output gate.

QUESTION 4

Convolutional LSTM(ConvLSTM) and Trajectory GRU (TrajGRU) are two deep learning architectures used for precipitation nowcasting. With respect to the said models identify the true statement from the given options:

Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

- A. The convolutional structure of ConvLSTM helps it to perform better than TrajGRU in nowcasting the precipitation.
- B. Both ConvLSTM and TrajGRU are location-variant in nature.
- C. The Trajectory GRU model can actively learn the location-variant structure for recurrent connections.
- D. TrajGRU is a traditional optical flow-based method used for nowcasting.

Answer: C

Detailed Solution:

The Trajectory GRU model can actively learn the location-variant structure for recurrent connections.

QUESTION 5

Researchers of the Earth Science Laboratory have collected a set of remote sensing data for the Indian subcontinent. They have trained and tested a deep neural network with this data. Later they tried to transfer the knowledge learned by the deep neural network on the source task to a new related target task. Which of the following method they have used in this task?

- A. Classification
- B. Domain Adaptation
- C. Regression
- D. Image Segmentation

Answer: B

Detailed Solution:

Deep domain adaptation allows us to transfer the knowledge learned by a particular deep neural network on a source task to a new related target task.

Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

QUESTION 6

Which of the following statements is/are true for the Triplet Adversarial Domain Adaptation (TriADA) method?

[Ref: Yan L, Fan B, Liu H, Huo C, Xiang S, Pan C. Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*. 2019 Dec 30; 58(5):3558-73.]

- (i) TriADA is used for Pixel-level classification of very high resolution (VHR) remote sensing images.
- (ii) TriADA method jointly considers both the source and target domains to learn a domain-invariant classifier by a novel domain similarity discriminator.

- A. Only (i)
- B. Only (ii)
- C. Both (i) and (ii)
- D. Neither (i) nor (ii)

Answer: C

Detailed Solution:

Ref: Yan L, Fan B, Liu H, Huo C, Xiang S, Pan C. Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*. 2019 Dec 30; 58(5):3558-73.

QUESTION 7

Which of the following statements indicates the objectives of earth system modeling?

- (i) Understanding the interactions between different components of earth system.
- (ii) Understanding future scenarios like climate change.
- (iii) Estimating the occurrence of specific events like floods and earthquakes.

Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (i) and (iii)
- (D) (i), (ii) and (iii)

Answer: D

Detailed Solution:

Aim of earth system modeling:

- (i) Understanding the interactions between different components of earth system
- (ii) Understanding future scenarios like climate change.
- (iii) Estimating the occurrence of specific events like floods and earthquakes.

QUESTION 8

In a typical physics-informed neural network (PINN) for solving a time-dependent partial differential equation, where are the core physics constraints most directly enforced during training?

- A. By constraining the optimization algorithm to search only in parameter regions allowed by the physics
- B. Only at the input layer by encoding physical variables as features
- C. In the loss function through residuals of the governing differential equations evaluated at collocation points
- D. In the final output layer by post-processing the network predictions to satisfy conservation laws.

Answer: C

Detailed Solution:

Physics guided loss function measures the deviations of the predictions of a Neural Network from the laws of physics.

Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

QUESTION 9

Identify the True statement/statements from the statements given below.

- (i) Process-based Model is a mathematical model for the entire process including all variables.
- (ii) Statistical Models tries to reproduce only the observable part of the process irrespective of the physics behind it.

- A. Only (i) is true.
- B. Both (i) and (ii) are true
- C. Only (ii) is true
- D. Both (i) and (ii) are false

Answer: B

Detailed Solution:

Both (i) and (ii) are true

QUESTION 10

Fill in the blanks with reference to the article "*Mitra A. Bayesian approach to Spatio-temporally Consistent Simulation of Daily Monsoon Rainfall over India. In Proceedings of the 25th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems 2017 Nov 7 (pp. 1-4).*"

During parameter estimation from observed data, the authors have used spatio-temporal smoothing using _____ so that the parameters learnt are spatially and temporally coherent.



Course Name: Machine Learning for Earth System Sciences

Week 07 - Assignment 07 - (Jul-2026)

- A. Markov Random Field
- B. Non parametric spatial clustering
- C. Chinese Restaurant Process
- D. LSTM

Answer: A

Detailed Solution:

Ref: "Mitra A. Bayesian approach to Spatio-temporally Consistent Simulation of Daily Monsoon Rainfall over India. In Proceedings of the 25th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems 2017 Nov 7 (pp. 1-4)."